Certified Prime Extensions and Uniform Coefficient Bounds

Philemon's totient conjecture, exact certificates, and a Euclidean prime generator

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Abstract

We identify the giant-prime computations in manuscripts [129] and [130] of the Verifiable Mathematical Reasoning Corpus, pin their source snapshot, and independently replay the seven-node certificate chain ending at the 18,886-digit prime r. A new search gives

$$P = 2347\,2^{8192}r + 1,$$

a 21,356-digit prime. Exact modular exponentiation and a greatest common divisor calculation certify this integer by Pocklington's theorem; the primality of the old corpus endpoint is an actual dependency of this proof.

We prove Philemon's original totient implication for every coefficient $1 \le a \le 64$, with no upper bound on m, using an explicit weighted factor-count obstruction and a finite classification of possible coefficient patterns. An explicit exceptional set has at most $\lfloor X/10 \rfloor$ members among the first X coefficients; outside this set, the original implication holds for every admissible m. Thus the proved coefficient set has lower density at least $9/10$. The unrestricted conjecture is not settled. We also investigate Euclidean prime generators. A corollary of the theorems of Booker–Pomerance and Munsch–Shparlinski proves eventual uniqueness of the new prime factor in OEIS A359504, while an exact finite computation checks its early terms. Neither this eventual statement nor the giant-prime computation is a full resolution of that original conjecture. We make no claim of a world record, historical first discovery, new general primality algorithm, or solution satisfying all six requested open-problem success gates. The complete primality verifier is embedded in this single source file.

Keywords. Certified prime generation; Pocklington; Proth; Euclidean prime generators; square-free smooth integers; reproducible computation.

1 Findings, provenance, and scope

The starting corpus was read at Git commit

`c30d40119134a59eda6fa5cc6331f27c8e67d034`.

The commit timestamp is 6 November 2026, 16:24:52 UTC. The repository's own README identifies it as a collection of research drafts, not peer-reviewed or formally verified results. Accordingly, descriptions in the old files were treated as source claims until the relevant certificate was replayed. The exact sources are listed in the references [1, 2].

Source	Explicitly certified endpoint	Decimal digits
Corpus [129]	$s = 5936p_5 + 1$	18,885
Corpus [130]	$r = 62988p_5 + 1$	18,886
This computation	$P = 2347\,2^{8192}r + 1$	21,356

Table 1: Constructed endpoints in the audited experiments. The two old endpoints s and r are siblings; both have parent p_5. These are not world-record claims.

The exact definition of p_5 is given in Section 3. The new integer P improves the number of decimal digits in the audited constructed endpoint by $21,356 - 18,886 = 2,470$, approximately 13.08%. This comparison concerns the completed certificate extensions in [129] and [130]. It excludes the much larger pre-existing Mersenne prime used as a benchmark input in [130]; P is not larger than that input. The phrase "new" in this report means newly found and certified in the present computation. It does not assert that no person has previously considered the same integer.

Corpus [129] also records an interrupted search among $2ms^2 + 1$, $1 \le m \le 10^6$, of approximately 37,775 digits. Its durable log summary contains 2,813 completed Fermat rejections and no certified prime from that search. The exact tested parameters are not included in the retrieved source, so their number cannot be used to skip the first 2,813 parameters in a future search. The present search uses a different family and retains its own machine-readable journals.

The work has two distinct outcomes. Extending the completed corpus certificate chain has a new unconditional answer. The request to solve an original named open problem under gates G1–G6 does not have a completed answer here. Section 10 records the precise distinction.

2 The classical certificate machinery

The following results are included to make the computational proof self-contained. They are classical; neither changing the coefficient nor reusing a previously certified factor creates a new primality theorem [3, 4, 5].

Theorem 2.1 (Pocklington certificate)**.** *Let $N > 1$, and let $F \mid N - 1$ be a completely factored positive integer with $F^2 > N$. For every distinct prime $q \mid F$, suppose that an integer a_q satisfies*

$$a_q^{N-1} \equiv 1 \pmod{N}, \qquad \gcd\left(a_q^{(N-1)/q} - 1, N\right) = 1. \tag{1}$$

Then N is prime.

Proof. Let ℓ be any prime divisor of N. The first congruence implies that a_q is a unit modulo ℓ and its order divides $N - 1$. The gcd condition means that this order does not divide $(N - 1)/q$. Therefore its q-adic valuation is exactly $v_q(N - 1)$; in particular $q^{v_q(F)} \mid \ell - 1$. Applying this reasoning for every prime dividing F gives $F \mid \ell - 1$. Thus $\ell > F > \sqrt{N}$. A composite integer has a prime divisor at most its square root, a contradiction. □

Corollary 2.2 (One known prime factor)**.** *Let q be a proven prime and $N = cq+1$ with $q^2 > N$. If*

$$a^{N-1} \equiv 1 \pmod{N}, \qquad \gcd(a^c - 1, N) = 1,$$

then N is prime. No factorization of c is needed.

Repeated factors must be counted with their full multiplicity. For $N = kq^m + 1$, one may take $F = q^m$ and use the reduced exponent kq^{m-1} in (1). In particular, for a quadratic lift the reduced exponent is kq, not k. Arbitrarily increasing m is possible at the level of the certificate; it does not guarantee that the resulting candidate is prime or that the search is inexpensive.

Theorem 2.3 (Proth certificate)**.** *Let K be positive and odd, $n \geq 1$, and $K < 2^n$. If $N = K2^n+1$ and*

$$a^{(N-1)/2} \equiv -1 \pmod{N}, \tag{2}$$

then N is prime.

Proof. Squaring the congruence gives the full Fermat congruence. Since N is odd, the gcd of the reduced power minus one with N is $\gcd(-2, N) = 1$. Take $F = 2^n$ in Theorem 2.1; the strict bound $K < 2^n$ implies $F^2 > N$. □

3 Replay of the existing corpus chain

Define

$$p_0 = 14^{512} - 14^{256} + 1, \tag{3}$$

$$p_i = k_i p_{i-1}^2 + 1 \quad (1 \leq i \leq 5), \tag{4}$$

$$(k_1, k_2, k_3, k_4, k_5) = (438, 13696, 5494, 13258, 22330),$$

$$r = 62988p_5 + 1. \tag{5}$$

The certificate starts only from the elementary primes 2 and 7, which the embedded verifier proves by trial division.

Node	Digits	Bits	Factored part F	Prime factor : base
p_0	587	1,950	$2^{256}7^{256}$	$2 : 11,\ 7 : 2$
p_1	1,177	3,908	p_0^2	$p_0 : 2$
p_2	2,357	7,829	p_1^2	$p_1 : 2$
p_3	4,718	15,670	p_2^2	$p_2 : 2$
p_4	9,439	31,354	p_3^2	$p_3 : 2$
p_5	18,882	62,722	p_4^2	$p_4 : 2$
r	18,886	62,738	p_5	$p_5 : 2$

Table 2: All seven rows were independently replayed in this computation. Every required full-power residue and gcd equals 1.

For the seed, put $x = 14^{256}$. Then $p_0 - 1 = x(x-1)$ and $x^2 - p_0 = x - 1 > 0$, so the known factors $2^{256}7^{256}$ suffice. For each quadratic child, the verifier checks the exact bound $p_{i-1}^4 > p_i$. For the final linear child it checks $p_5^2 > r$. Every factor used in a later certificate must already belong to the verifier's set of proven primes.

The canonical SHA-256 of r, computed from its decimal ASCII digits without a trailing newline, is

```
ca4c8532ec289a68d0fe9769f87f3797aa560f9f1a141866da17b917c11c1081
```

This agrees with the fingerprint in corpus [130]. The hash checks integer identity; primality comes from the order argument and arithmetic checks.

4 A 21,356-digit prime using the old certificate

Theorem 4.1 (Newly certified direct extension)**.** *With r defined by (3)–(5), the integer*

$$\boxed{P = 2347\,2^{8192}r + 1} \tag{6}$$

is prime. It has $21,356$ *decimal digits and* $70,941$ *bits.*

Proof. The replay in Section 3 proves that r is prime. The exact computations recorded below and reproduced by the embedded verifier give

$$r^2 > P, \tag{7}$$

$$2^{P-1} \bmod P = 1, \tag{8}$$

$$\gcd(2^{2347\,2^{8192}} - 1, P) = 1. \tag{9}$$

Corollary 2.2 with $q = r$ and $c = 2347\,2^{8192}$ proves the claim. The digit count and bit length are computed from the reconstructed integer, rather than estimated from floating-point logarithms. □

4.1 Exact identity and arithmetic replay

The decimal SHA-256 is

```
6f4975bbc4efe37c474e15c24dcf3950b130a0338cbb03220550730ad73d4ad8
```

The first and last 100 decimal digits are:

```
First 100:
1900934091840816417588332438892074719851779737904 5
0536936694995805642794694993342213793081072030845 7
Last 100:
3765210359410730303576822527729752395049010946925 8
7857784089425869718764917890275526163526035754188 9
```

The direct replay used gmpy2 2.3.2 with GMP 6.3.0. On this host, while a separate search was running, the full power in (8) took 20.1567 seconds; the reduced power and gcd in (9) took 2.9247 seconds. These are observed wall times, not complexity bounds or performance guarantees on another machine.

The complete decimal expansion is included in the companion verification archive. It is unnecessary for the definition: equations (3)–(6) specify every digit uniquely, and the single-file verifier reconstructs it without downloading any data.

5 Search design and retained evidence

The direct continuation considered

$$L_k = k\,r\,2^{8192} + 1, \qquad k \in \{1, 3, 5, \ldots, 1,999,999\}.$$

There were exactly 10^6 coefficients before sieving. For each odd prime $\ell \le 10^8$, let $C \equiv r2^{8192} \pmod{\ell}$. If $C = 0$, that prime cannot divide any L_k. Otherwise the unique prohibited class is

$$k \equiv -C^{-1} \pmod{\ell}.$$

Writing $k = k_0 + 2j$ gives the equivalent index condition

$$j \equiv (-C^{-1} - k_0)2^{-1} \pmod{\ell}.$$

This is an exact sieve of a linear family. It tests $5,761,455$ primes including 2 and leaves $60,985$ coefficients. The parity check is automatic because every candidate is odd.

Three single-thread workers tested survivors using OpenPFGW 4.1.8, with GWNUM 31.5, in base-two probable-prime mode [6]. The survivor $k = 2347$ has zero-based sorted index 67 in the retained candidate list. It passed after 43.3 seconds of the parallel run. At that moment 67 tests had completed; after the already-running tests finished, the durable journals held 69 completed tests: one pass and 68 software rejections. The process ended at 44.1472 seconds. The completion order of parallel workers is not the numerical ordering of the coefficients.

The search result was only a screening pass. It became the prime in Theorem 4.1 after the separate exact computations (7)–(9). The final proof does not rely on floating-point FFT roundoff behavior or on the program printing "PRP". The software rejection records describe what that search did; they are not needed for the primality proof or a claim of a globally first prime.

An earlier, separate Proth search used $K2^{131072}+1$ with $K = 202611080000001 + 2j$, $0 \le j < 10^6$. It was stopped after 285 completed software rejections, with no passing candidate. Its journals are retained separately from the next search.

A second Proth search used $K2^{131113} + 1$ with $K = 202611 + 2j$, $0 \le j < 10^6$. Sieving through 10^8 left 60,875 coefficients. Seven single-thread workers used the OpenPFGW $N - 1$ test. The execution ended without a terminal summary. Reconciliation of the durable journals finds 3,429 completed tests, all software rejections, with no passing or unrecognized result. The last periodic checkpoint recorded 3,425 completions after 1463.1 seconds; later completed records are included in the reconciled count. Unlogged candidates are not classified. This interrupted search supplies no additional prime theorem.

The sieve was checked on independent small-range enumerations of both families. The downloaded OpenPFGW archive matched the publisher's SHA-256,

```
aadf885e2d6489866bb7eeea2ba3f75a3fa02e679f1d6a86fd2632e042a99c99
```

This provenance check is separate from the mathematical certificate. The archive accompanying this report preserves the search program, sieve source, parameter files, journals, and exact verification output. It does not redistribute the third-party executables.

6 A bounded audit of the much larger old candidates

Corpus [130] also discusses

$$q = 2^{136279841} - 1, \qquad U = 28q^2 + 1, \qquad V = 66q^3 + 1.$$

The source gives $82,048,642$ and $123,072,962$ decimal digits, respectively. These were candidate integers, not certified successes. In particular, benchmarking a modular square did not evaluate the complete Fermat congruences that their proposed certificates would require.

The present additional audit checks every prime $\ell \le 10^9$ by computing $q_\ell = (2^{136279841} - 1) \bmod \ell$ and testing

$$28q_\ell^2 + 1 \pmod{\ell}, \qquad 66q_\ell^3 + 1 \pmod{\ell}.$$

No prime in that range divides either candidate. The run covered $50,847,534$ primes, including the parity check, and took 8.77406 seconds on one core. The complete bounded-audit code is reproduced in Appendix B. Products stay below the 64-bit unsigned limit when the enforced bound is at most 10^9.

This is exactly a finite small-factor exclusion. It does not establish that U or V is prime, and it does not change the certified maximum in Table 1. No primality assumption about q is needed for this modular audit.

7 An original named problem close to the certificate method

The most directly related named target located in the review is Conjecture 4.1 of Philemon [7]:

> Let $n = am + 1$, where $a \geq 1$, $m > 1$, and $p = P^-(m)$ is the least prime divisor of m. If $a < p$ and $m \mid \varphi(n)$, then n is prime.

This formulation predates the present work and retains the original quantifiers. The author's version 5, dated 10 April 2021, explicitly states it as a conjecture. The same paper proves the prime-power case $m = p^e$ earlier. Original online discussions contain additional partial arguments, including a case with two distinct prime factors in m [8, 9]. No established unrestricted resolution was located in the sources examined; that search result is not a proof of historical novelty.

7.1 Why the giant-prime certificate does not settle it

Pocklington's gcd condition works separately at every prime divisor ℓ of n. It forces a sufficiently large factor of the order modulo *each* ℓ, and thereby rules out a small divisor. The aggregate condition $m \mid \varphi(n)$ is different: factors of m may be supplied by different terms in

$$\varphi(n) = \prod_{\ell^e \| n} \ell^{e-1}(\ell - 1).$$

For example, an order condition that is merely nontrivial modulo n does not automatically make its reduced-power gcd equal to one. Replacing the original totient condition by the local gcd conditions would change the problem and would invoke an existing theorem.

An elementary necessary condition illustrates the remaining difficulty. If a composite counterexample existed, put $t = \varphi(n)/m$. Then

$$1 \leq t < a < p, \qquad \gcd(n, m) = 1.$$

For every prime $u \mid m$, the inequality $t < p \leq u$ implies $v_u(\varphi(n)) = v_u(m)$. This constrains the allocation of large prime factors among the terms $\ell - 1$, but does not force all of them into a single term. Such valuation reductions already occur in the original discussion; they are not claimed as new results here.

8 A factor-count obstruction for Philemon's conjecture

Philemon's Conjecture 4.1 asks whether

$$m > 1, \qquad n = am + 1, \qquad 1 \leq a < p := \mathrm{lpf}(m), \qquad m \mid \varphi(n) \quad \Longrightarrow \quad n \text{ is prime.} \tag{10}$$

Here $\mathrm{lpf}(m)$ is the least prime divisor of m. The prime-power case and a two-prime case precede the present work [7, 9]. The factor allocation and product-expansion strategy used below also

have precedents in the 2020 discussion [9], including the answers by Haran and user140242. We include the full justification because the distinction between $\varphi(n)$ and the orders modulo individual prime factors is essential. The weighted tail estimate and final factor-count bound are derived here, without a claim of historical priority. The argument does not settle the unrestricted implication (10).

Lemma 8.1 (Exact allocation in a hypothetical counterexample)**.** *Suppose the hypotheses in* (10) *hold and n is composite. Then m is odd. There are integers $C \geq 1$, $s \geq 1$, $c_i \geq 2$, and $m_i \geq p$ such that*

$$n = C\prod_{i=1}^{s}(c_i m_i + 1), \qquad \prod_{i=1}^{s} m_i = m, \qquad U := C\prod_{i=1}^{s} c_i < a. \tag{11}$$

Every c_i is even. The numbers $c_i m_i + 1$ are precisely the distinct prime factors of n greater than p, each with multiplicity one; every prime factor of C is less than p. In particular, $C < a$ and $U \geq C2^s$. Moreover,

$$a - U + \frac{1}{m} = U\left\{\prod_{i=1}^{s}\left(1 + \frac{1}{c_i m_i}\right) - 1\right\} < \frac{Us}{2p - s}. \tag{12}$$

Proof. Write $\varphi(n) = bm$. Since n is composite, $\varphi(n) < n - 1 = am$, and hence $1 \leq b < a < p$. Thus $\gcd(b, m) = 1$. Also $\gcd(n, m) = 1$ because $n = am + 1$. If m were even, then $p = 2$ and $a = 1$, contradicting $b < a$. Therefore m is odd and $p \geq 3$.

Call a prime divisor ℓ of n active if $\gcd(\ell - 1, m) > 1$. Any active prime exceeds p. If any prime $\ell > p$ occurred in n with multiplicity at least two, then $\ell \mid \varphi(n)$ and $\ell \nmid m$ would give $\ell \mid b$, contradicting $b < p < \ell$. If a prime $\ell > p$ were inactive, then $\ell - 1 \mid \varphi(n)$ and $\gcd(\ell - 1, m) = 1$ would give $\ell - 1 \mid b$, again impossible. No prime divisor of n equals p, since $\gcd(n, m) = 1$. Consequently, the active primes are exactly the prime divisors of n greater than p, all with multiplicity one. List them as $\ell_1, \ldots, \ell_s$ and put $C = n/(\ell_1 \cdots \ell_s)$. All prime factors of C are less than p, so $\gcd(\varphi(C), m) = 1$. The condition $m > 1$ and $m \mid \varphi(n)$ ensures $s \geq 1$.

For each i define the full part of $\ell_i - 1$ supported on the prime divisors of m by

$$m_i = \prod_{t \mid m,\ t \text{ prime}} t^{v_t(\ell_i - 1)}, \qquad c_i = \frac{\ell_i - 1}{m_i}.$$

Because $\gcd(b, m) = \gcd(\varphi(C), m) = 1$ and $\varphi(n) = \varphi(C)\prod_i(\ell_i - 1) = bm$, the t-adic valuations on both sides show that $\prod_i m_i = m$ exactly. Every active m_i is a nontrivial divisor supported on primes at least p, so $m_i \geq p$. Each m_i is odd, whereas $\ell_i - 1$ is even; therefore $c_i \geq 2$ is even.

Expanding $C\prod_i(c_i m_i + 1)$ gives $n = Um + \Delta$, where $\Delta > 1$ in the composite case. Indeed, if $s \geq 2$ the nonleading positive terms already sum to more than one. If $s = 1$, then $\Delta = C$; the alternative $C = 1$ would make $n = \ell_1$ prime. Comparison with $n = am + 1$ now gives $(a - U)m = \Delta - 1 > 0$, proving $U < a$ and $C < a$.

Dividing the product formula for n by m proves the equality in (12). Since $c_i m_i \geq 2p$,

$$\prod_i\left(1 + \frac{1}{c_i m_i}\right) \leq \left(1 + \frac{1}{2p}\right)^s < \frac{1}{1 - s/(2p)}.$$

The last inequality follows by comparing the binomial expansion with the geometric series, using $\binom{s}{j} \leq s^j$. Its denominator is positive: $p > a > U \geq 2^s$ implies $s < 2p$. Subtracting one and multiplying by U proves the strict inequality in (12). □

Theorem 8.2 (Parity-sensitive factor-count obstruction)**.** *In every composite counterexample to* (10)*, the notation in Lemma 8.1 satisfies the stronger weighted bound*

$$a - U < \sum_{i=1}^{s} \frac{1}{c_i} \le \frac{s}{2}. \tag{13}$$

In particular,

$$2(a-U) < s, \qquad \textit{equivalently} \qquad s \ge 2(a-U)+1. \tag{14}$$

Consequently, a composite counterexample must have at least five distinct prime factors greater than p if n is odd, and at least three such factors if n is even.

Proof. Use Lemma 8.1 and let $h = a - U$ and $S = \sum_i 1/c_i$. Since $S \le s/2 < a < p$, all the following denominators are positive. The elementary symmetric terms in $\prod_i (1 + 1/(c_i p))$ are bounded by the corresponding powers of S/p, so comparison with a geometric series gives

$$\prod_i \left(1 + \frac{1}{c_i m_i}\right) \le \prod_i \left(1 + \frac{1}{c_i p}\right) < \frac{1}{1 - S/p}.$$

The equality in (12), followed by $p > a$, yields

$$h < \frac{US}{p - S} < \frac{US}{a - S} = \frac{(a-h)S}{a - S}.$$

Multiplication by $a - S$ and cancellation give $ah < aS$, and hence $h < S \le s/2$. This proves both (13) and (14).

The integer U is even. If n is odd, then a is even because m is odd; thus h is a positive even integer, so $h \ge 2$ and $s \ge 5$. If n is even, then a and h are odd; thus $h \ge 1$ and $s \ge 3$. □

Corollary 8.3. *Writing $\Omega(m)$ for the number of prime factors of m counted with multiplicity, a composite counterexample has $\Omega(m) \ge 5$ when n is odd and $\Omega(m) \ge 3$ when n is even. In the notation of Lemma 8.1, the corresponding lower bounds are $a \ge 32C + 2$ and $a \ge 8C + 1$, respectively.*

Proof. Each $m_i > 1$ and $\prod_i m_i = m$, so $\Omega(m) \ge s$. Also $a = U + h \ge C2^s + h$; apply the theorem and the parity bounds on h. □

Corollary 8.4 (All coefficients up to 32)**.** *The original implication* (10) *is true for every integer $1 \le a \le 32$, with no bound on m.*

Proof. Suppose instead that a composite counterexample has $a \le 32$. If n is odd, the preceding corollary gives $a \ge 32C + 2 \ge 34$, a contradiction. If n is even, then C is even and $s \ge 3$. Since $U = C \prod_i c_i < a \le 32$, the alternatives $s \ge 4$, $C \ge 4$, or any $c_i \ge 4$ are all impossible: each would give $U \ge 32$. Therefore $C = 2$, $s = 3$, $c_1 = c_2 = c_3 = 2$, and $U = 16$. The strict bound $2(a - U) < s = 3$ forces $a - U = 1$, so $a = 17$.

Now $p > 17$ and every prime factor of every m_i is at least p; thus $3 \nmid m_i$. The number $\ell_i = 2m_i + 1$ is a prime greater than p, so it is not divisible by 3. These two facts force $m_i \equiv 2 \pmod 3$ and $\ell_i \equiv 2 \pmod 3$ for all three indices. Consequently the product formula for n gives $n = 2\ell_1\ell_2\ell_3 \equiv 1 \pmod 3$, whereas $n = 17m + 1 = 17m_1m_2m_3 + 1 \equiv 2 \pmod 3$. This contradiction proves the claim. No primality assumption on the individual m_i was used. □

8.1 A uniform coefficient range through 64

Corollary 8.5 (All coefficients up to 64)**.** *The original implication* (10) *holds for every integer* $1 \le a \le 64$*, with no upper bound on* m*.*

Proof. Suppose, to the contrary, that n is composite and $a \le 64$. Use the exact decomposition in Lemma 8.1, and put $h = a - U$. Thus $U < a \le 64$, $U \ge C2^s$, all c_i are even, and (14) gives $2h < s$. The integers m_i are odd, and every prime divisor of every m_i is at least $p > a$. Distinct active primes with the same coefficient $c_i = c_j$ have $m_i \ne m_j$.

Complete coefficient classification. If n is odd, then C is odd, h is positive and even, and $s \ge 5$. The inequality $2^s \le U < 64$ forces $s = 5$. Now $U = C\prod_i c_i$ is a positive multiple of 32 below 64, so $U = 32$, $C = 1$, and all five c_i equal 2. The inequality $2h < 5$ forces $h = 2$, hence $a = 34$.

If n is even, then C is even, h is positive and odd, and $s \ge 3$. Here $2^{s+1} \le U < 64$, so $s = 3$ or $s = 4$. For $s = 4$, U is a positive multiple of 32 below 64; therefore $U = 32$, $C = 2$, all $c_i = 2$, $h = 1$, and $a = 33$. For $s = 3$, U is a positive multiple of 16 below 64, and $2h < 3$ gives $h = 1$. Thus (U, a) is one of $(16, 17)$, $(32, 33)$, and $(48, 49)$. Factoring

$$\frac{U}{16} = \frac{C}{2}\prod_{i=1}^{3}\frac{c_i}{2} \in \{1, 2, 3\}$$

gives precisely the following cases, up to permutation of the c_i:

a	C	s	$(c_1, \ldots, c_s)$	U	h
17	2	3	$(2, 2, 2)$	16	1
33	2	3	$(2, 2, 4)$	32	1
33	4	3	$(2, 2, 2)$	32	1
33	2	4	$(2, 2, 2, 2)$	32	1
34	1	5	$(2, 2, 2, 2, 2)$	32	2
49	2	3	$(2, 2, 6)$	48	1
49	6	3	$(2, 2, 2)$	48	1

(15)

In particular, this classification includes both the three-factor case $C = 4$ and the four-factor case $C = 2$ at $a = 33$.

Four exclusions modulo 3. In every row where all $c_i = 2$, one has $p > 17$, hence $3 \nmid m_i$. Since $2m_i + 1$ is prime and exceeds p, it is not divisible by 3. Consequently $m_i \equiv 2 \pmod 3$ and $2m_i + 1 \equiv 2 \pmod 3$. The two formulas for n therefore require

$$C2^s \equiv a2^s + 1 \pmod 3. \tag{16}$$

For $(a, C, s) = (17, 2, 3), (33, 4, 3), (33, 2, 4), (34, 1, 5)$, the left and right residues are, respectively, $(1, 2), (2, 1), (2, 1), (2, 0)$. These four rows are impossible. The row $(49, 6, 3)$ survives this particular congruence and is treated separately below.

Monotone bounds for the remaining rows. For positive coordinates t_i define

$$R_U(\boldsymbol{c}; \boldsymbol{t}) = U\left\{\prod_{i=1}^{s}\left(1 + \frac{1}{c_i t_i}\right) - 1\right\}.$$

This expression decreases in each coordinate t_i. The equality in (12) says

$$h + \frac{1}{m} = R_U(\boldsymbol{c}; \boldsymbol{m}). \tag{17}$$

All remaining rows have $h = 1$, so any coordinatewise upper bound on the right side that is less than 1 gives a contradiction.

For $a = 33$, $C = 2$, and $\boldsymbol{c} = (2, 2, 4)$, one has $p \geq 37$. Order the first two coordinates so that $m_1 < m_2$. The value $m_1 = 37$ would give the composite active factor 75. Every integer from 38 through 40 has a prime factor below 37, so $m_1 \geq 41$; oddness and distinctness then give $m_2 \geq 43$. Also $m_3 \geq 37$. Hence

$$R_{32}((2, 2, 4); \boldsymbol{m}) \leq R_{32}((2, 2, 4); (41, 43, 37)) = \frac{64466}{65231} < 1. \tag{18}$$

Both remaining rows have $a = 49$ and therefore $p \geq 53$. An integer $m_i < 53^2$ supported on primes at least 53 must itself be prime. Among integers d in the interval $[53, 113]$ for which both d and $2d + 1$ are prime, the possibilities are exactly

$$53, \quad 83, \quad 89, \quad 113. \tag{19}$$

This small enumeration is elementary: the other primes in the interval and proper factors of $2d + 1$ are listed here:

d	59	61	67	71	73	79
proper factor of $2d + 1$	7	3	5	11	3	3

d	97	101	103	107	109
proper factor of $2d + 1$	3	7	3	5	3

The four retained numbers $107, 167, 179, 227$ are prime by trial division through their square roots. Integers $d \leq 113$ that are not prime cannot be any m_i, because their least prime factors are below 53.

For $C = 2$ and $\boldsymbol{c} = (2, 2, 6)$, order the two coordinates with coefficient 2. The preceding enumeration gives $m_1 \geq 53$, $m_2 \geq 83$, and $m_3 \geq 53$. Consequently

$$R_{48}((2, 2, 6); \boldsymbol{m}) \leq R_{48}((2, 2, 6); (53, 83, 53)) = \frac{209366}{233147} < 1. \tag{20}$$

Finally, consider $C = 6$ and $\boldsymbol{c} = (2, 2, 2)$. Order all three coordinates increasingly. If they are not exactly $(53, 83, 89)$, their coordinatewise lower bounds are $(53, 83, 113)$: the first three eligible values below 113 are precisely those in the exceptional triple. Thus in every nonexceptional case,

$$R_{48}((2, 2, 2); \boldsymbol{m}) \leq R_{48}((2, 2, 2); (53, 83, 113)) = \frac{477402}{497087} < 1. \tag{21}$$

The exceptional triple gives, exactly,

$$m = 53 \cdot 83 \cdot 89 = 391511, \qquad n = 6 \cdot 107 \cdot 167 \cdot 179 = 19191306.$$

But

$$n - 1 - 49m = 19191305 - 19184039 = 7266 \neq 0, \tag{22}$$

so it does not satisfy the required identity $n = 49m + 1$. Every row of (15) is excluded, proving the corollary. □

The bound in Corollary 8.5 is on the coefficient a, not on the size of m or n. The argument therefore covers infinitely many possible inputs in each of the 64 coefficient classes. It remains a partial result for (10); neither the remaining coefficients nor historical priority are settled here.

8.2 A coefficient set of lower density at least 9/10

The quantitative factor-count obstruction also proves the original implication for an explicit infinite set of coefficients. Define, for each positive integer h,

$$M_h = \begin{cases} 2^{2h+1}, & h \text{ even}, \\ 2^{2h+2}, & h \text{ odd}, \end{cases} \qquad \mathcal{E}_h = \{h + tM_h : t = 1, 2, 3, \ldots\}, \qquad \mathcal{E} = \bigcup_{h \geq 1} \mathcal{E}_h.$$

The positive-multiple condition $t \geq 1$ is part of the definition. In particular, the residue representative $a = h$ is not included merely by belonging to that residue class.

Theorem 8.6 (Uniform coefficient-density bound)**.** *For every coefficient $a \notin \mathcal{E}$, the original implication (10) holds for every $m > 1$ satisfying $a < \mathrm{lpf}(m)$. For each positive integer X,*

$$\#(\mathcal{E} \cap \{1, \ldots, X\}) \leq \left\lfloor \frac{X}{10} \right\rfloor. \tag{23}$$

Consequently, at least $X - \lfloor X/10 \rfloor$ coefficient classes among $1, \ldots, X$ satisfy the conjectured implication uniformly in m. The explicit set $\mathbb{Z}_{\geq 1} \setminus \mathcal{E}$ has lower asymptotic density at least $9/10$.

Proof. Suppose a composite counterexample exists, and retain the notation in Lemma 8.1. Put $h = a - U > 0$. Theorem 8.2 gives $s \geq 2h + 1$. Since every c_i is even, $2^s \mid U$.

If a is even, then n and C are odd, while h is even. It follows that

$$2^{2h+1} \mid U = a - h.$$

If a is odd, then n and C are even, while h is odd. The additional factor of two in C gives

$$2^{2h+2} \mid U = a - h.$$

Thus in either case $M_h \mid a - h$. Because $U = a - h > 0$, the quotient is a positive integer; therefore $a \in \mathcal{E}_h \subseteq \mathcal{E}$. Contraposition proves the first assertion, with all quantifiers over m unrestricted.

For a fixed h and X, the number of terms in $\mathcal{E}_h$ up to X is

$$\max\left\{0, \left\lfloor \frac{X - h}{M_h} \right\rfloor\right\} \leq \frac{X}{M_h}.$$

Only finitely many such sets meet $\{1, \ldots, X\}$, and their union has size at most the sum of their sizes. Summing the convergent bounds over all h gives

$$\begin{aligned} \#(\mathcal{E} \cap \{1, \ldots, X\}) &\leq X \left(\sum_{j=1}^{\infty} 2^{-(4j+1)} + \sum_{j=0}^{\infty} 2^{-(4j+4)} \right) \\ &= X \left(\frac{1}{30} + \frac{1}{15} \right) = \frac{X}{10}. \end{aligned}$$

The left side is an integer, giving (23). Taking a lower limit of the complementary proportions proves the density assertion. □

Membership in $\mathcal{E}$ is a finite calculation for any specified a: a relevant h must satisfy $h + M_h \leq a$. The set $\mathcal{E}$ contains only coefficients not certified by this density argument; membership does not assert the existence of a counterexample. Corollary 8.5 additionally settles every coefficient up to 64, including those that lie in $\mathcal{E}$. Theorem 8.6 concerns coefficient classes, not the frequency of primes or a proportion of arbitrary pairs (a, m). The unrestricted conjecture remains unresolved by these arguments.

8.3 Restricted unconditional primality certificates

The preceding results give an unconditional instance of the certificate proposed conditionally in Philemon's Theorem 4.2 [7]. The order argument itself is the earlier one; the point is the explicit coefficient range on which its premise has now been established here.

Corollary 8.7 (An unconditional instance of the extended Lucas criterion)**.** *Let $m > 1$, $1 \le a < \mathrm{lpf}(m)$, and $N = am + 1$. Assume either $a \le 64$ or $a \notin \mathcal{E}$. Suppose that for every prime $q \mid m$ there is an integer b_q such that*

$$b_q^{N-1} \equiv 1 \pmod{N}, \qquad b_q^{(N-1)/q} \not\equiv 1 \pmod{N}. \tag{24}$$

Then N is prime.

Proof. The first congruence makes b_q a unit modulo N. Let $d_q = \mathrm{ord}_N(b_q)$. Then $d_q \mid N-1$ while $d_q \nmid (N-1)/q$. Because $a < \mathrm{lpf}(m)$, one has $\gcd(a, m) = 1$. If $q^e \| m$, it follows that $q^e \| N - 1$ and $q^e \mid d_q \mid \varphi(N)$. Combining these divisibilities for the distinct primes dividing m gives $m \mid \varphi(N)$. Apply Corollary 8.5 or Theorem 8.6, as appropriate. □

The noncongruence in (24) is taken modulo N itself. This corollary is restricted to its stated coefficient set; it does not justify omitting a Pocklington gcd check for arbitrary inputs. No general speed advantage or historical priority for a new primality criterion is claimed.

No proof in this report closes that allocation problem in general. Consequently, Philemon's conjecture cannot be used as an assumption in the new-prime proofs. The certificates in Sections 2 and 4 are unconditional without it.

9 A named conjecture on a greedy Euclidean prime sequence

This section records a separate theoretical consequence and an exact bounded experiment. The theoretical result is explicitly a corollary of existing theorems on smooth squarefree representatives. It does not establish the all-step assertion in the original conjecture, and it does not depend on the size of the certified prime constructed elsewhere in this paper.

9.1 The original sequence and conjecture

Joel Brennan's OEIS entry A359504, dated January 3, 2023, defines the following sequence [10]. Put $S_0 = \varnothing$. For a finite set S of primes, write

$$\mathcal{D}(S) = \left\{ \prod_{q \in T} q : T \subseteq S \right\},$$

where the empty product is 1. Thus $\mathcal{D}(S)$ consists of the squarefree positive integers whose prime factors belong to S. Recursively, define

$$\begin{aligned} P_{n+1} &= \min\{d \in \mathcal{D}(S_n) : d+1 \text{ has a prime divisor outside } S_n\}, \\ a_{n+1} &= \min\{q : q \text{ is prime},\ q \mid P_{n+1}+1,\ q \notin S_n\}, \\ S_{n+1} &= S_n \cup \{a_{n+1}\}. \end{aligned}$$

These definitions preserve the original order of the two minimizations: first minimize the product, and then select its smallest new prime factor. They do not minimize the new prime across all available products.

The construction continues indefinitely. Indeed, the product of all primes in S_n is an eligible product, since every prime factor of that product plus one lies outside S_n. Each term is a new prime. The first four terms are $2, 3, 7, 5$, with selected products $1, 2, 6, 14$ respectively.

Brennan's conjecture. At every step, $P_{n+1}+1$ has exactly one distinct prime divisor outside S_n. The original entry states "at most one"; eligibility supplies "at least one," so these formulations are equivalent. Multiplicity of the new factor is not restricted by the original assertion.

9.2 Two published inputs

A positive integer is p-smooth when every prime factor is at most p. Booker and Pomerance [11, Theorem 1] prove that, for every prime $p \notin \{5, 7\}$, every residue class modulo p has a positive squarefree p-smooth representative. Applied to the residue -1, such a representative has all its prime factors strictly below p.

Munsch and Shparlinski [12, Theorem 1.4] prove the stronger asymptotic size bound

$$M(p) \le p^{3/2+o(1)},$$

where $M(p)$ is the smallest bound permitting squarefree p-smooth representatives for all residue classes. In particular, for each fixed $\varepsilon > 0$, every sufficiently large prime p admits such a representative d of -1 with $d \le p^{3/2+\varepsilon}$.

Booker and Pomerance's Corollary 7 concerns a different selection rule: it minimizes the new prime over all products. The argument below transfers their representative theorem to Brennan's product-first rule.

Corollary 9.1 (Every prime occurs)**.** *The sequence A359504 contains every prime exactly once.*

Proof. Distinctness follows from the definition. Suppose that a prime never occurs, and let p be the least such prime. The displayed initial terms give $p \ge 11$, handling both exceptional moduli in the published theorem. All primes smaller than p occur, so there is an index N after which all of them belong to S_n.

Choose a squarefree integer d with all prime factors below p and $d \equiv -1 \pmod{p}$, using Booker–Pomerance. For every $n \ge N$, the integer d belongs to $\mathcal{D}(S_n)$ and remains eligible, because $p \mid d+1$ and $p \notin S_n$. Therefore $P_{n+1} \le d$ and $a_{n+1} \le d+1$ for every $n \ge N$. This would give infinitely many distinct primes bounded by the fixed integer $d+1$, a contradiction. □

Let μ_n denote the least prime outside S_n. The first corollary implies $\mu_n \to \infty$: each fixed finite set of primes is eventually contained in S_n.

Corollary 9.2 (Eventual uniqueness, with multiplicity one)**.** *For each $\varepsilon > 0$, there is an index N_ε such that*

$$P_{n+1} \le \mu_n^{3/2+\varepsilon}, \qquad a_{n+1} \le \mu_n^{3/2+\varepsilon} + 1 \quad (n \ge N_\varepsilon).$$

Consequently, for all sufficiently large n, the integer $P_{n+1}+1$ has exactly one prime divisor outside S_n, and that prime occurs to exponent one.

Proof. For sufficiently large n, Munsch–Shparlinski provides a squarefree μ_n-smooth integer $d_n \equiv -1 \pmod{\mu_n}$ with $d_n \le \mu_n^{3/2+\varepsilon}$. Since $\mu_n \nmid d_n$, all prime factors of d_n are below μ_n and hence belong to S_n. Thus d_n is eligible, and the two bounds follow from the selection rule.

Fix $0 < \varepsilon < 1/2$. As $\mu_n \to \infty$, these bounds give $P_{n+1}+1 < \mu_n^2$ eventually. Every prime outside S_n is at least μ_n. If $P_{n+1}+1$ had two distinct new prime divisors, or if a new prime divisor occurred at least twice, it would be at least μ_n^2. Both possibilities are excluded. Eligibility guarantees the existence of a new prime divisor, completing the proof. □

The proof also gives an exact local criterion: at any step, an eligible squarefree integer $d \equiv -1 \pmod{\mu_n}$ with prime factors below μ_n and $d+1 < \mu_n^2$ certifies uniqueness and exponent one at that step. This criterion is sufficient; its failure is not a counterexample.

9.3 Exact finite experiment and its scope

We implemented a smallest-prime-factor sieve and an availability queue for all squarefree products $d \leq B$. For each such product, a counter stores how many of its distinct prime factors have not yet appeared. Adding a new prime decrements these counters on the corresponding squarefree multiples. Products whose counters reach zero enter a min-heap. A popped product is factored after adding one, and the distinct factors outside the current prime set are counted.

This handles decreases in the selected products, such as the transition from 14 to 10 after the first four terms. At every successful step, the least eligible product in the bounded queue is also the least eligible product globally, because every omitted product exceeds B. Already inspected products cannot acquire a new absent factor when the known-prime set grows. These invariants justify the resulting sequence prefix; exhaustion of the bounded queue is only a stopping condition for the experiment.

With $B = 100{,}000{,}000$, the run generated the first 3,795,778 sequence terms and inspected 58,574,992 available squarefree subset products. No selected product plus one had two distinct new prime factors. The largest generated prime was 99,999,971. The recorded running time was 22.75 seconds in the execution environment used for this paper; this is an observation rather than a hardware-independent performance claim. The first 5,000 terms were compared against the complete official OEIS b-file, with no mismatches; the common 5,000th term is 77,291. The finite search checks distinct new factors, which is the property in Brennan's conjecture.

9.4 The remaining logical gap

The eventual-uniqueness corollary proves that any failures of Brennan's conjecture occur among finitely many steps. The proof does not supply a numerical index after which failures are impossible. The finite experiment cannot cover an unspecified exceptional range. Combining the asymptotic theorem with the displayed search therefore does not prove the original assertion for every step.

For the requested success conditions, the target retains its published definition, but the established statement has the weaker quantifier "all sufficiently late steps." Thus it does not satisfy the required full-scope G2/G4 closure. The theoretical argument is a corollary of published representation bounds, so it supplies no independent claim of the methodological novelty demanded by G5. No priority or G6 claim is made. The explicit outcomes are the two corollaries and the bounded verification, with the original all-step conjecture left unresolved.

10 Success-gate audit and remaining conclusions

Gate	Required success criterion	Result of this investigation
G1	A previously named problem with a prior original statement	Philemon Conjecture 4.1 and OEIS A359504 are concrete prior targets. A new large prime by itself is not such a solved conjecture.
G2	Preserve the original proposition and quantifiers	The targets are stated unchanged. The coefficient restrictions and eventual uniqueness are explicit partial results; neither replaces an original unrestricted quantifier.
G3	The target was actually unresolved	The primary sources examined still state the targets conjecturally. No claim is made that a search can rule out all unpublished or overlooked prior work.

Gate	Required success criterion	Result of this investigation
G4	An unconditional complete proof of the original target	Complete proofs are supplied for the specific prime certificates, $a \leq 64$, and the explicit coefficient set of lower density at least 9/10, all uniformly in m. Not achieved for either unrestricted original conjecture. The unspecified eventual cutoff in A359504 is not closed.
G5	A substantive new argument, not a re-expression	The prime tests are classical. The weighted factor-count estimate works for a partial range of Philemon's conjecture; A359504 uses existing analytic theorems. No new method resolving an entire target is claimed.
G6	No prior identical result or essentially identical proof	No historical-first claim is justified for the integers, algorithms, or analytic corollaries in this report.

The request to extend the completed corpus chain is fully answered. The partial theorems do not meet G4 for either original conjecture, and the historical priority required by G6 has not been established.

Reproducibility and attribution

This report was prepared with assistant-supported source review, mathematical analysis, programming, and exact computation. Classical theorems are attributed to their sources. The original corpus is licensed CC BY 4.0; its seed definitions and certificate parameters are reused with attribution to manuscripts [129] and [130]. The verifier below is self-contained: the complete source needs no external bibliography, integer file, image, or custom style to compile. A normal TeX installation with the packages in the preamble suffices.

A Complete exact-integer primality verifier

The program reconstructs the complete dependency chain, proves the small base primes by trial division, and checks every certificate explicitly. It uses gmpy2 when available; otherwise Python's exact integers supply the same mathematical operations with a slower runtime. The checks are ordinary conditional statements and remain active under Python's optimization flag. No probable-prime oracle is used.

```
"""Exact primality proofs. Python 3; optional gmpy2 accelerates the same checks.
No input files, prime tables, PRP oracle, or network are used.
"""
import hashlib
import json
import math
import sys
import time

if hasattr(sys, "set_int_max_str_digits"):
    sys.set_int_max_str_digits(0)
try:
    import gmpy2
    Z, modpow, gcd = gmpy2.mpz, gmpy2.powmod, gmpy2.gcd
    backend = "gmpy2 " + gmpy2.version()
except ImportError:
    Z, modpow, gcd = int, pow, math.gcd
    backend = "Python standard-library exact integers"

def need(condition, explanation):
    if not condition:
        raise ArithmeticError(explanation)

```

```
known = set()
for q in (2, 7):
    need(all(q % d for d in range(2, math.isqrt(q) + 1)),
         "base-prime trial division failed")
    known.add(Z(q))

def fingerprint(label, n, expected_digits, expected_bits,
                expected_hash=None):
    decimal = str(n)
    digest = hashlib.sha256(decimal.encode("ascii")).hexdigest()
    need(len(decimal) == expected_digits, label + ": digit count")
    need(n.bit_length() == expected_bits, label + ": bit length")
    if expected_hash is not None:
        need(digest == expected_hash, label + ": decimal identity")
    print(json.dumps({"label": label, "status": "CERTIFIED",
          "digits": len(decimal), "bits": n.bit_length(),
          "sha256_decimal": digest}), flush=True)
    return n

def pocklington(label, n, factors):
    need(n > 2 and n % 2, label + ": odd positive integer")
    F = Z(1)
    distinct = set()
    for q, exponent, a in factors:
        need(q in known, label + ": unknown factor primality")
        need(exponent >= 1 and q not in distinct,
             label + ": invalid factor list")
        distinct.add(q)
        F *= q**exponent
    need((n-1) % F == 0, label + ": F does not divide n-1")
    need(F*F > n, label + ": insufficient factored part")
    for q, exponent, a in factors:
        need(1 < a < n, label + ": witness range")
        need(modpow(Z(a), n-1, n) == 1,
             label + ": Fermat congruence failed")
        t = modpow(Z(a), (n-1)//q, n)
        need(gcd(t-1, n) == 1, label + ": gcd failed")
    known.add(n)
    return n

started = time.perf_counter()
x = Z(14)**256
p = pocklington("p0", x*x-x+1,
                [(Z(2), 256, 11), (Z(7), 256, 2)])
fingerprint("p0", p, 587, 1950)
sizes = [(1177,3908), (2357,7829), (4718,15670),
         (9439,31354), (18882,62722)]
for i, (k, size) in enumerate(zip(
        (438,13696,5494,13258,22330), sizes), 1):
    parent = p
    p = pocklington("p"+str(i), k*parent*parent+1,
                    [(parent,2,2)])
    fingerprint("p"+str(i), p, *size)
r = pocklington("r", 62988*p+1, [(p,1,2)])
fingerprint("r", r, 18886, 62738,
    "ca4c8532ec289a68d0fe9769f87f3797"
    "aa560f9f1a141866da17b917c11c1081")
P = pocklington("P", 2347*(Z(2)**8192)*r+1, [(r,1,2)])
fingerprint("P", P, 21356, 70941,
    "6f4975bbc4efe37c474e15c24dcf3950b"
    "130a0338cbb03220550730ad73d4ad8")



print("ALL CERTIFICATES PASSED;", backend,
      "elapsed seconds", round(time.perf_counter()-started, 3))
```

B Source for the bounded giant-candidate audit

Compile with `g++ -O3 -std=c++17 audit.cpp -o audit` and run `./audit 1000000000`. At 2, both integers are odd; all remaining primes are generated by an Eratosthenes sieve. Modular exponentiation uses only exact integer operations.

```
#include <cstdint>
#include <cstdlib>
#include <iostream>
#include <vector>
#include <chrono>

static std::uint64_t power_mod(std::uint64_t a, std::uint64_t e, std::uint64_t m) {
    std::uint64_t r=1;
    while(e) { if(e&1) r=r*a%m; e>>=1; if(e) a=a*a%m; }
    return r;
}

int main(int argc,char**argv) {
    const std::uint64_t bound=argc>1 ? std::strtoull(argv[1],nullptr,10) : 100000000;
    if(bound>1000000000ULL) return 2;
    auto begin=std::chrono::steady_clock::now();
    std::vector<bool> composite(bound/2+1,false);
    for(std::uint64_t p=3;p*p<=bound;p+=2) if(!composite[p/2])
        for(std::uint64_t n=p*p;n<=bound;n+=2*p) composite[n/2]=true;
    std::uint64_t count=1,hits2=0,hits3=0;
    for(std::uint64_t p=3;p<=bound;p+=2) if(!composite[p/2]) {
        ++count;
        std::uint64_t a=power_mod(2,136279841,p);
        std::uint64_t q=(a+p-1)%p;
        std::uint64_t sq=q*q%p;
        if((28*sq+1)%p==0) { ++hits2; std::cout << "N2_FACTOR " << p << " q_mod " << q << std::endl; }
        if((66*(sq*q%p)+1)%p==0) { ++hits3; std::cout << "N3_FACTOR " << p << " q_mod " << q << std::
     endl; }
    }
    double dt=std::chrono::duration<double>(std::chrono::steady_clock::now()-begin).count();
    std::cout << "bound="<<bound<<" primes_checked="<<count<<" N2_hits="<<hits2<<" N3_hits="<<hits3<<"
     seconds="<<dt<<std::endl;
}
```

C Source for the exact A359504 finite search

This program maintains the original priority rule, including products smaller than the previously selected product that become eligible after a new prime is inserted. Its invariant and the precise scope of the finite result are explained in the main text. The plus-sign run is the one used for the A359504 claim. The optional minus-sign mode is retained for reproducibility of the exploratory code; no general theorem about that mode is asserted here.

```
#include <algorithm>
#include <chrono>
#include <cstdint>
#include <fstream>
#include <functional>
#include <iostream>
#include <queue>
#include <string>
#include <vector>
using namespace std;
int main(int argc,char**argv){
 const int M=argc>1?stoi(argv[1]):1000000;
 const int sign=argc>2?stoi(argv[2]):1;
 auto start=chrono::steady_clock::now();
 vector<int> spf(M+2),ps;
 for(int i=2;i<=M+1;i++){
  if(!spf[i]){spf[i]=i;ps.push_back(i);}
```

```
  for(int p:ps){if(p>spf[i]||1LL*p*i>M+1)break;spf[p*i]=p;}
 }
 vector<int8_t> missing(M+1,0);
 for(int i=2;i<=M;i++){
  int p=spf[i],v=i/p;
  missing[i]=(v%p==0||missing[v]<0)?-1:missing[v]+1;
 }
 vector<bool> seen(M+2,false);
 priority_queue<int,vector<int>,greater<int>> heap;
 heap.push(1);
 int n=0,largest=0,maxP=0,examined=0; bool failure=false;
 ofstream prefix(string("research/named_problem/prefix_")+(sign==1?"plus":"minus")+".csv");
 prefix<<"n,P,new_prime\n";
 auto add=[&](int q,int P){
  n++; largest=max(largest,q);seen[q]=true;
  if(n<=10000)prefix<<n<<","<<P<<","<<q<<"\n";
  if(n<=59)cout<<q<<((n==59)?"\n":" ");
  for(int x=q;x<=M;x+=q){if(missing[x]>=0){missing[x]--;if(missing[x]==0)heap.push(x);}}
 };
 if(sign==-1){add(2,0);add(3,0);}
 while(!heap.empty()){
  int P=heap.top();heap.pop();examined++;maxP=max(maxP,P);
  int x=P+sign;
  vector<int> unseen;
  while(x>1){int p=spf[x]; if(!seen[p])unseen.push_back(p); while(x%p==0)x/=p;}
  if(unseen.size()>1){cout<<"COUNTEREXAMPLE step="<<n+1<<" P="<<P<<" factors=";for(int p:unseen)cout<<p
     <<",";cout<<"\n";failure=true;break;}
  if(unseen.size()==1)add(unseen[0],P);
 }
 double elapsed=chrono::duration<double>(chrono::steady_clock::now()-start).count();
 cout<<"RESULT M="<<M<<" sign="<<sign<<" terms="<<n<<" largest_prime="<<largest<<" max_P="<<maxP<<"
    products="<<examined<<" failure="<<failure<<" elapsed_seconds="<<elapsed<<"\n";
 return 0;
}
```

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